RAI-818: INTELLIGENT MOBILE ROBOTICS

Potential Field Approach

- **Potential Functions**

Text : Principles of Robot Motion – Theory, Algorithms and Implementations
H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

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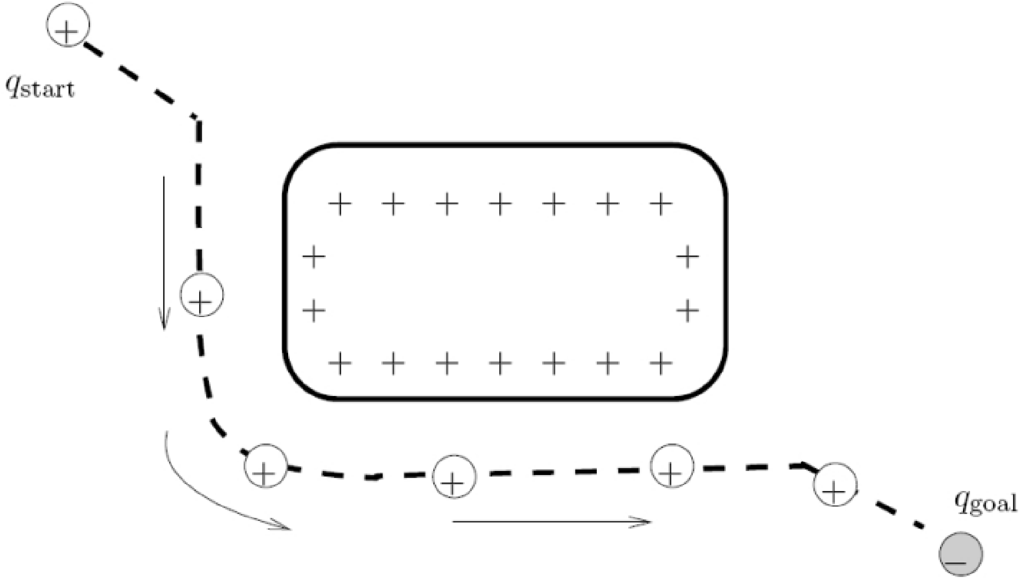
Potential Fields

The Basic Idea

- A really simple idea:
  - Suppose the goal is a point ***g∈ℜ2***
  - Suppose the robot is a point ***r ∈ℜ2***
  - Think of a ***spring*** drawing the robot toward the goal and away from obstacles
  - Can also think



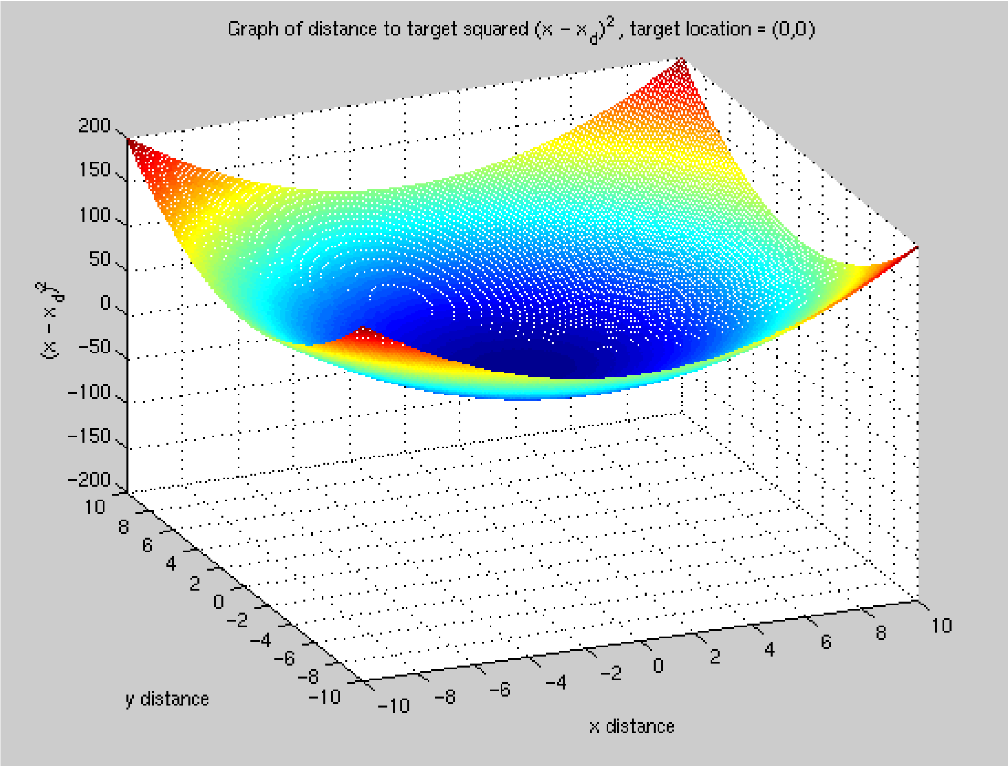
Defining futures

Potential Fields

- **Potential Fields**
  - **Basic Idea**

Another Idea

- **Think of the goal as the *bottom of a bowl***
- **The robot is at the rim of the bowl**
- **What will happen?**



Potential Fields

- Potential Fields
  - Basic Idea

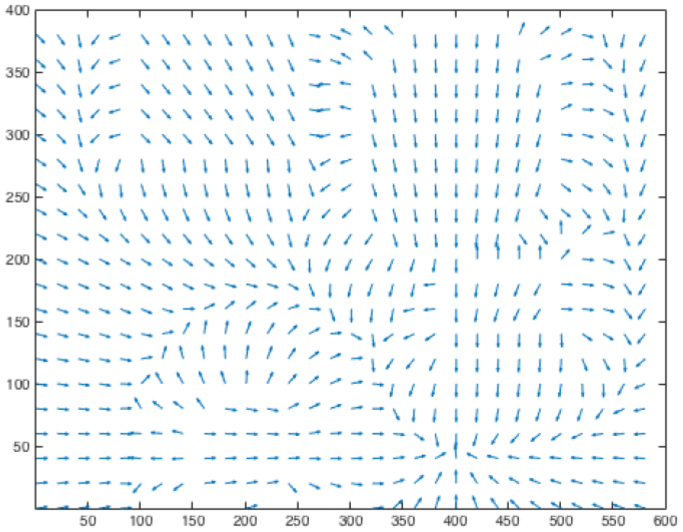
Another Idea

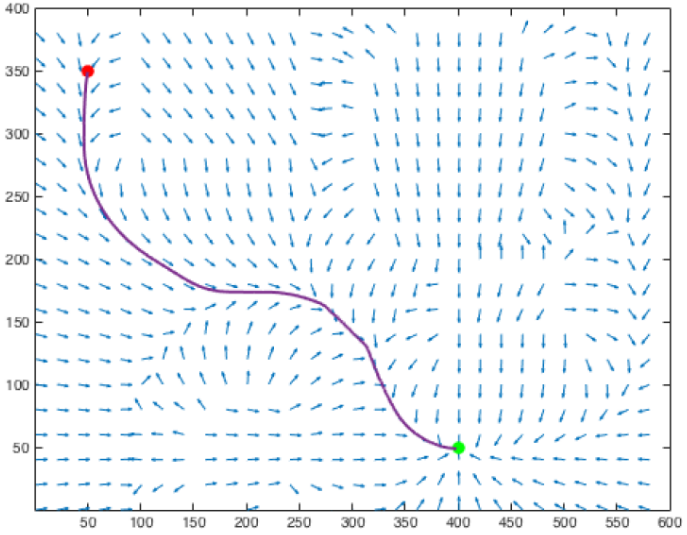
- **Both the bowl and the spring analogies are ways of storing *potential energy***
- **The robot *moves to a lower energy* configuration**
- **A *potential function is a function $U : \Re^m \rightarrow \Re$***
- **Energy is minimized by following *negated gradient* of the *potential energy* function:**

$$\text{gradient at } q \in \mathbb{R}^n :$$

$$\nabla U(q) = \left[\frac{\partial U}{\partial q_1}(q), \ldots, \frac{\partial U}{\partial q_n}(q) \right]^T$$

- **We can now think of a *vector field over the space of all q's ...***
  - **at every point in time, the robot looks at the vector at that point and goes in the vector direction**





Attractive/Repulsive Potential Field

- ***Desired Objectives***
  - **robot moves toward the goal (*attractive* potential)**
  - **robot stays away from the obstacles (*repulsive* potential)**

$$U(q) = U_{att}(q) + U_{rep}(q)$$

U_{att} is the *attractive* potential - move to the goal

U_{rep} is the *repulsive* potential - avoid obstacles

Defining futures

Potential Fields

- **Potential Fields**
  - Basic Idea
  - Attractive / Repulsive Field

Attractive/Repulsive Potential Field

- ***Desired Objectives***
  - robot moves toward the goal (*attractive potential*)
  - robot stays away from the obstacles (*repulsive potential*)

$$U(q) = U_{att}(q) + U_{rep}(q)$$

- ***Attractive Potential $U_{att}(q)$***
  - monotonically increasing with distance from q_{goal}
  - example: conic potential (scaled distance to goal, $\varsigma > 0$ scaling factor)
  - what's the gradient?

$$U_{att}(q) = \varsigma \|q, q_{goal}\| \qquad \nabla U_{att}(q) = \frac{\varsigma}{\|q, q_{goal}\|}(q - q_{goal})$$

  - what's the magnitude of the gradient at q

$$\|\nabla U_{att}(q)\| = \begin{cases} \varsigma & q \neq q_{goal} \\ undefined & q = q_{goal} \end{cases}$$

What happens at the goal?

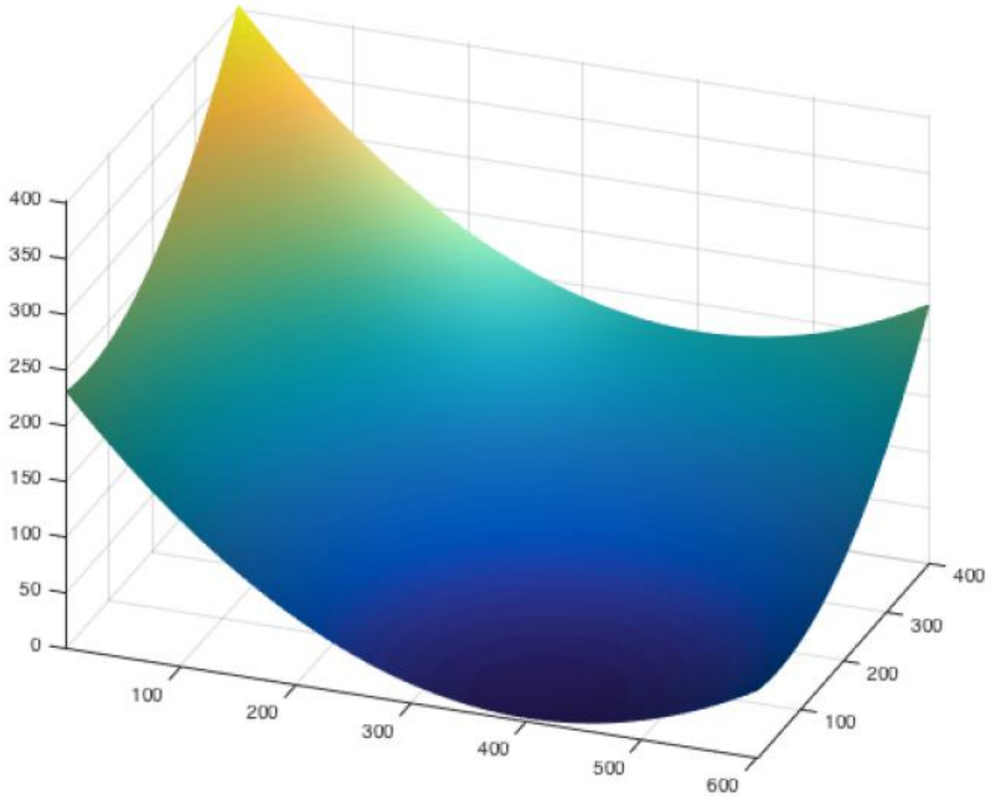
Defining futures

Potential Fields

- **Potential Fields**
  - Basic Idea
  - **Attractive / Repulsive Field**

Attractive/Repulsive Potential Field

$$U_{att}(q) = \varsigma \| q, q_{goal} \|$$



Defining futures

Attractive/Repulsive Potential Field

- ***Desired Objectives***
  - robot moves toward the goal (*attractive potential*)
  - robot stays away from the obstacles (*repulsive potential*)

$$U(q) = U_{att}(q) + U_{rep}(q)$$

- ***Attractive Potential*** $U_{att}(q)$
  - monotonically increasing with distance from q_{goal}
  - preference: continuously differentiable + magnitude decreases as robot approaches q_{goal}
  - example : quadratic potential ($\varsigma > 0$ scaling factor)

$$U_{att}(q) = \frac{1}{2}\varsigma\|q, q_{goal}\|^2$$

  - what's the gradient? $\nabla U_{att}(q) = \varsigma(q - q_{goal})$
  - what's the magnitude of the gradient at q

$$\|\nabla U_{att}(q)\| = \varsigma\|q, q_{goal}\|$$

What happens when robot is very far from goal?

Defining futures

Potential Fields

Attractive/Repulsive Potential Field

- ***Desired Objectives***
  - robot moves toward the goal (*attractive potential*)
  - robot stays away from the obstacles (*repulsive potential*)

$$U(q) = U_{att}(q) + U_{rep}(q)$$

- ***Attractive Potential $U_{att}(q)$***
  - monotonically increasing with distance from q_{goal}
  - preference: continuously differentiable + magnitude decreases as robot approaches q_{goal} does not produce very large velocities
  - example : combine conic and quadratic potentials ($\varsigma > 0$)

$$U_{att}(q) = \begin{cases} \frac{1}{2}\varsigma\|q, q_{goal}\|^2 & if\|q, q_{goal}\| \le d^*_{goal} \\ d^*_{goal}\varsigma\|q, q_{goal}\|^2 - \frac{1}{2}\varsigma(d^*_{goal})^2 & if\|q, q_{goal}\| > d^*_{goal} \end{cases}$$

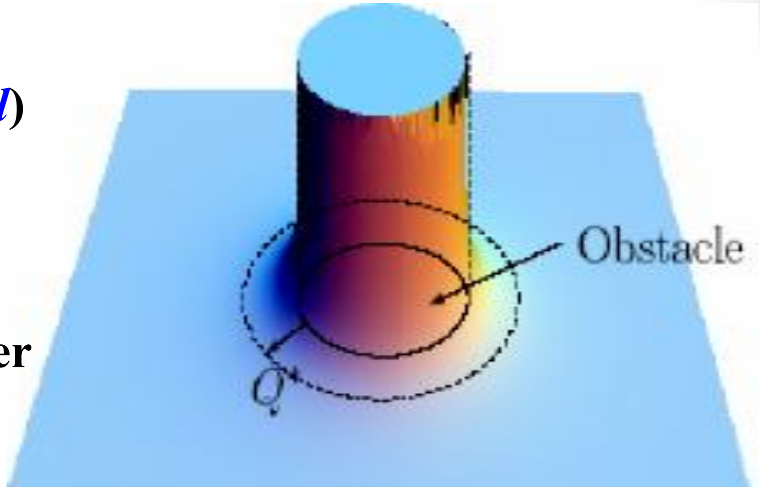
(d^*_{goal}: threshold from goal where planner switches between conic and quadratic potentials)

  - what's the gradient? $U_{att}(q) = \begin{cases} \varsigma(q - q_{goal}) & if\|q, q_{goal}\| \le d^*_{goal} \\ d^*_{goal}\,\varsigma(q - q_{goal}) / \|q, q_{goal}\| & if\|q, q_{goal}\| > d^*_{goal} \end{cases}$

Defining futures

Attractive/Repulsive Potential Field

- ***Desired Objectives***
  - robot moves toward the goal (*attractive potential*)
  - robot stays away from the obstacles (*repulsive potential*) $U(q) = U_{att}(q) + U_{rep}(q)$



- ***Repulsive Potential $U_{rep}(q)$***
  - the closer the robot is to an obstacle, the stronger the repulsive force should be

$$U_{rep}(q) = \begin{cases} \frac{1}{2}\eta\left(\frac{1}{D(q)} - \frac{1}{Q^*}\right)^2 & if\ D(q) \le Q^* \\ 0 & otherwise \end{cases}$$

$$\nabla U_{rep}(q) = \begin{cases} \eta\left(\frac{1}{D(q)} - \frac{1}{Q^*}\right)\frac{1}{(D(q))^2}\nabla D(q), & if\ D(q) \le Q^* \\ 0 & otherwise \end{cases}$$

  - *D(q)*: distance to the closest obstacle; *η > 0* scaling factor
  - *Q**: threshold to allow robot to ignore obstacles far away from it

What happens around points equidistant from obstacles?

Defining futures

- **Potential Fields**
  - Basic Idea
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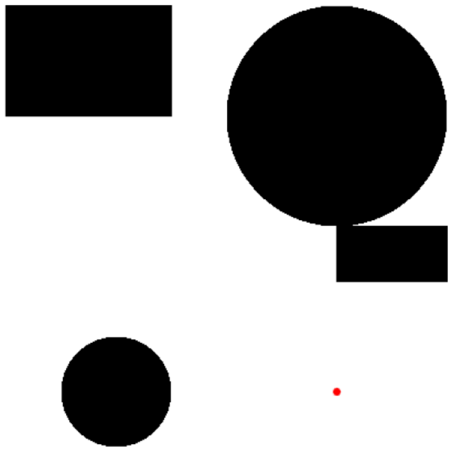
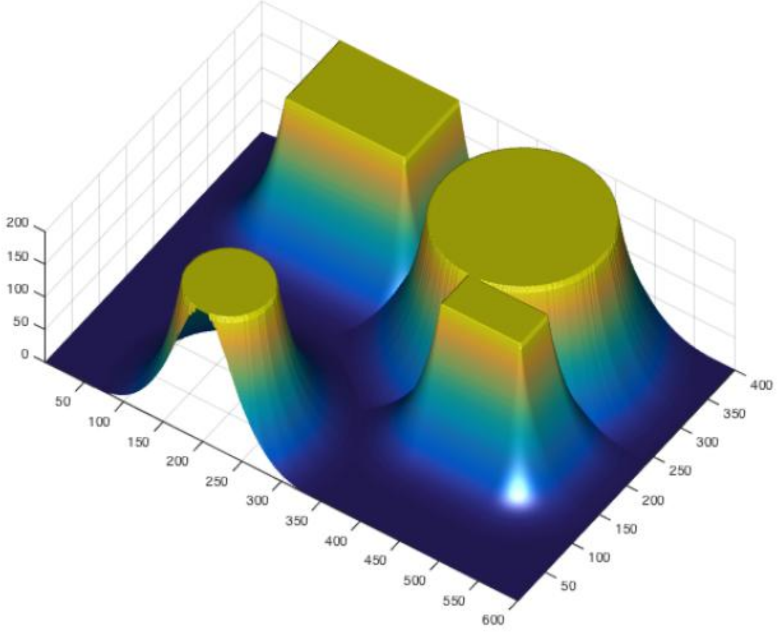
Attractive/Repulsive Potential Field

$$U_{rep}(q) = \begin{cases} {}^{1}/_{2}\,\eta\left(\dfrac{1}{D(q)} - \dfrac{1}{Q^*}\right)^2 & if\ \ D(q) \le Q^* \\ 0 & otherwise \end{cases}$$

$$\nabla U_{rep}(q) = \begin{cases} \eta\left(\dfrac{1}{D(q)} - \dfrac{1}{Q^*}\right)\dfrac{1}{(D(q))^2}\nabla D(q), & if\ \ D(q) \le Q^* \\ 0 & otherwise \end{cases}$$





Defining futures

- **Potential Fields**
  - Basic Idea
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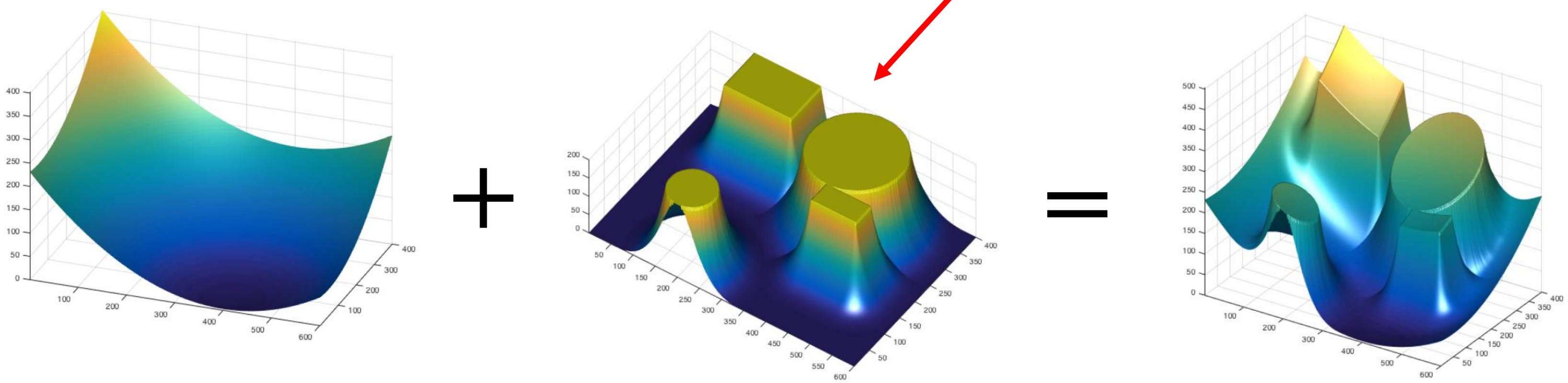
Attractive/Repulsive Potential Field

$$U(q) = U_{att}(q) + U_{rep}(q)$$

$$U_{att}(q) = \varsigma \| q, q_{goal} \|$$

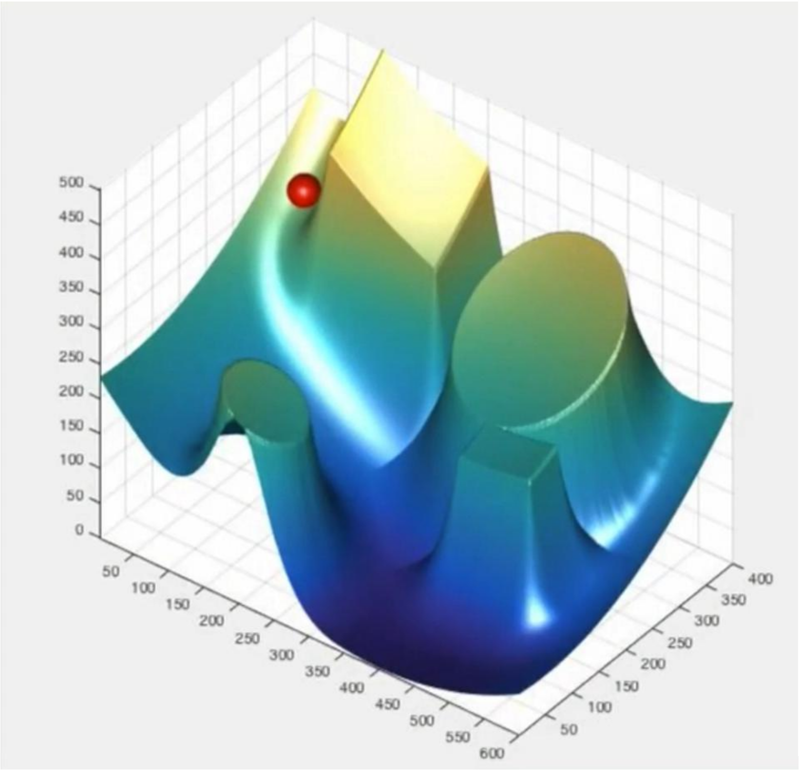
$$U_{rep}(q) = \begin{cases} {}^{1}/_{2}\, \eta \left(\dfrac{1}{D(q)} - \dfrac{1}{Q^*} \right)^2 & \text{if } D(q) \le Q^* \\ 0 & \text{otherwise} \end{cases}$$

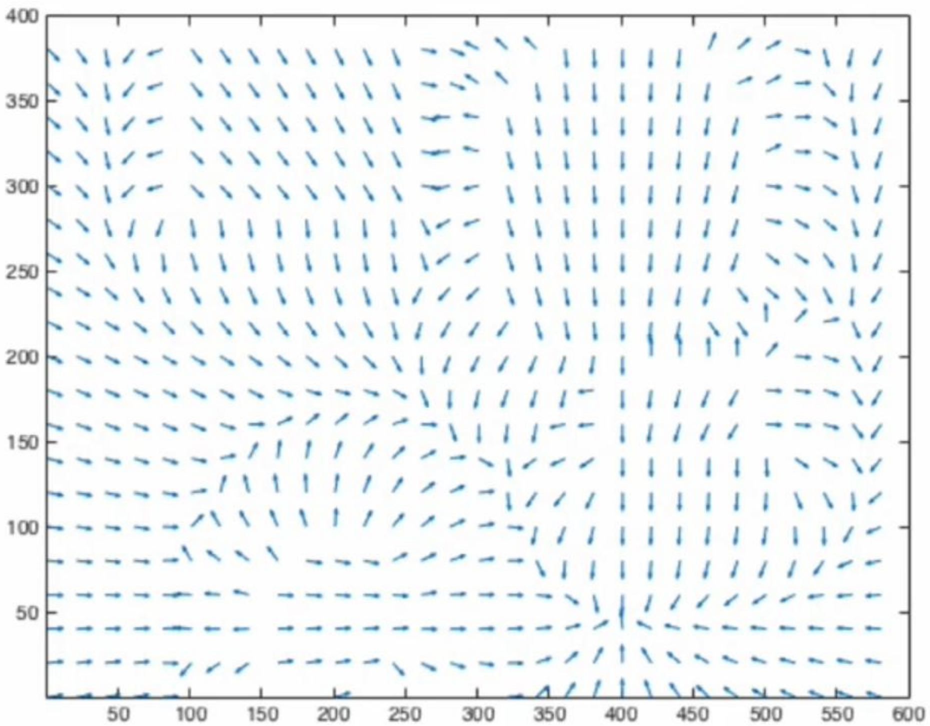


$$F(q) = -\nabla U(q)$$

Defining futures

Attractive/Repulsive Potential Field





Attractive/Repulsive Potential Field

- ***Desired Objectives***
  - robot moves toward the goal (*attractive potential*)
  - robot stays away from the obstacles (*repulsive potential*)

$$U(q) = U_{att}(q) + U_{rep}(q)$$

- ***Repulsive Potential $U_{rep}(q)$***
  - minimum distance to *i-th* obstacle $\quad d_i(q) = \min_{c\, \in Obstacle_i} (q, c)$
  - for convex obstacles (c is closest point to q) $\quad \nabla d_i(q) = \dfrac{(q - c)}{\|q, c\|}$
  - repulsive potential for each obstacle

$$U_{rep}(q) = \begin{cases} {}^1\!/_2\, \eta \left(\dfrac{1}{d_i(q)} - \dfrac{1}{Q^*} \right)^2 & if \;\; d_i(q) \le Q^* \\ 0 & otherwise \end{cases}$$

  - overall repulsive potential as sum of obstacle potentials

$$U_{rep}(q) = \sum_i U_{rep_i}(q)$$

Defining futures

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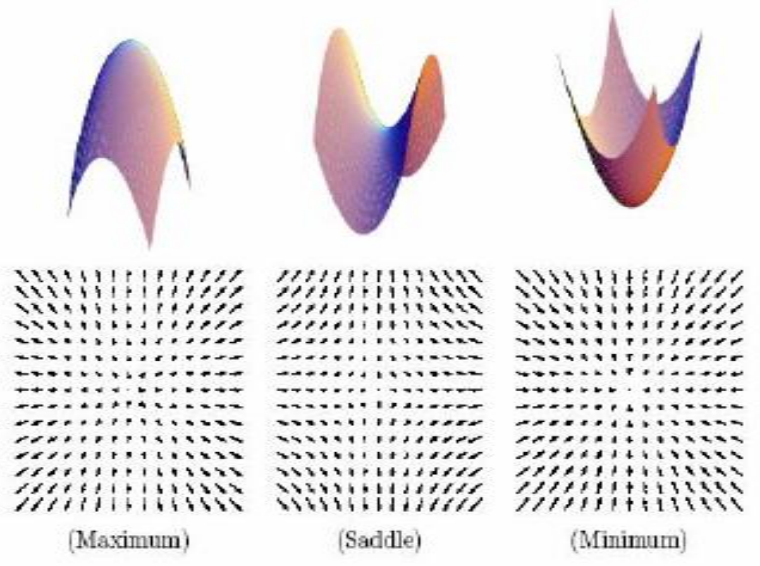
Gradient Descent

- A simple way to get to the bottom of a potential

$$\dot{c}(t) = -\nabla U(c(t))$$

Robot terminates the motion at a point where the gradient *vanishes* i.e., it has reached a critical point

- A critical point is a point *x s.t. q** where $\nabla U(q^*) = 0$
  - Max, min, saddle
  - Stability?



Potential Fields

The Hessian

- **For a *1-D* function, how do we know we are at a *unique minimum* (or maximum)?**

- **The *Hessian* is the m×m matrix of second derivatives**

- **If the Hessian is nonsingular ($Det(H) \neq 0$), the critical point is a unique point**
  - **if *H* is *positive definite* point is a minimum**
  - **if *H* is *negative definite*, a maximum**
  - **if *H* is *indefinite*, a saddle point**

Defining futures

Gradient Descent More Formally

Input

A means to compute a gradient $\nabla U(q)$ at a point q

Output

A sequence of points $[q(0), q(1), q(2),\ldots]$

1: $q(0) = q_{start}$
2: $i = 0$
3: **while** $\cancel{\nabla U(q(i)) \neq 0}$ **do** $\|\nabla U(q(i))\| < \varepsilon$
4: $\quad q(i+1) = q(i) + \alpha(i)\, \nabla U(q(i))$
5: $\quad i = i + 1$
6: **end while**

$q(i)$ is the value of q at the i^{th} iteration

value of $\alpha(i)$ determines the step size at the i^{th} iteration

Defining futures

Potential Fields

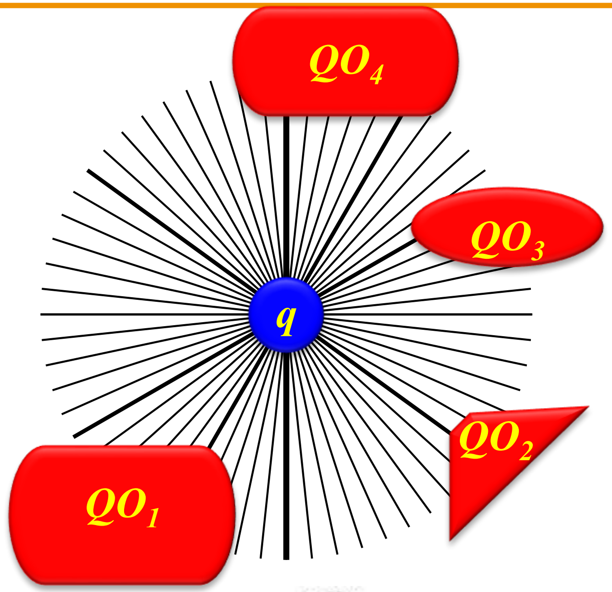
Distance Computation

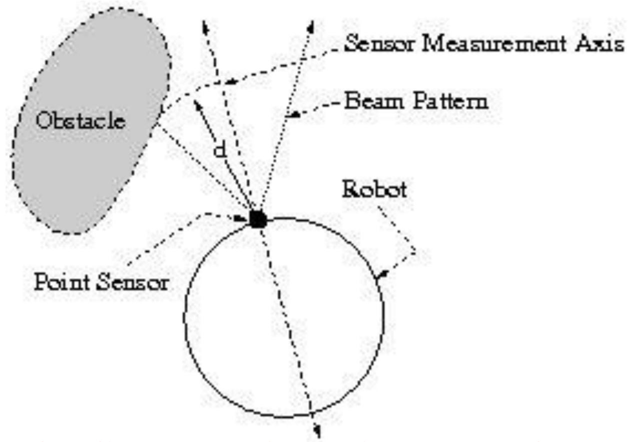
- We will now discuss some implementation issues in constructing the *attractive* /*repulsive* potential functions
- The *attractive* potential is pretty straight forward - if the robot knows its current location and the goal location
- The *repulsive* potential is a bit more challenging to compute since distance to obstacles is required
- The are generally 2 different methods to compute the distance to obstacles (and hence the repulsive function)
  - sensor based implementation - chapter 2 sensor concepts
  - grid based implementation – assumes the configuration space can be discretized into grid

Potential Fields

Distance Computation – Sensor Info

- Planar robot equipped with radially distributed sensors
- Range sensors approximate the value of the ***raw distance function***, ρ,
- $D(q)$ corresponds to the global minimum value of the ***raw distance function***, ρ,
- $d_i(q)$ corresponds to the local minimum with respect to θ of $\rho(q, \theta)$
- Any sensor whose reading is less than that of its left and right neighbours is a local minimum (such sensors face the closest point on their respective obstacles)
- The sensors point in the direction that maximally brings the robot closer to the obstacles i.e., $-\nabla d_i(q)$





Obstacle distance function may be incorrect for partial occlusions